

Chapter 10

Heat-Electricity Equivalence

10.1 Objectives

1. To demonstrate the relationship between electrical energy and heat energy.
2. To determine the numerical equivalence between joules and calories.

10.2 Tools

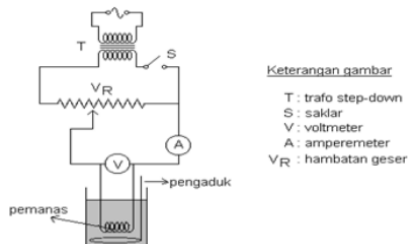


Figure 10.1: Schematic diagram of Apparatus

1. Calorimeter with stirrer and heater.
2. Voltmeter AC (0 – 15 V)
3. Amperemeter AC (0 – 5 A)
4. Thermometer (-10 – 110°C)
5. Step-Down Transformer (220 V to 18 V AC)
6. Variable resistor
7. Switch
8. Scale

9. Timer (clock)

10.3 Theoretical Foundation

The mechanical equivalent of heat (J) is the amount of energy required to produce one calorie of heat. The mechanical equivalent of heat was first measured by Joule by using the mechanical energy of a falling weight to stir water in a calorimeter, thereby increasing its temperature. Another method is by converting electrical energy into heat energy in a resistance wire immersed in water contained in a calorimeter. If E volts is the voltage across a conducting coil and I amperes is the current passing through the coil for t seconds, then the electrical energy in the coil becomes Eit . The number that expresses the amount of electrical energy equivalent to one unit of heat energy is called the electrical heat equivalent. The mechanical equivalent of heat was first measured by Joule by using the mechanical energy of a falling weight to stir water in a calorimeter, thus heating the water. Another method is by converting electrical energy into heat energy in a resistance wire immersed in water inside the calorimeter. If this energy is converted into heat H (calories), then the mechanical equivalent of heat J is:

$$J = \frac{Eit \text{ joule}}{H \text{ calories}} \quad (10.1)$$

If H is calculated using a calorimeter and its contents, where the temperature increases from $\theta_1^\circ\text{C}$ to $\theta_2^\circ\text{C}$ so that,

$$H = (M_s + W)(\theta_2 - \theta_1) \quad (10.2)$$

where M is the mass of the water in the calorimeter, s is the specific heat of water, and W is the water equivalent of the calorimeter and the stirrer. W can be calculated from the mass

and the specific heat of the calorimeter and the stirrer. From equations (10.1) and (10.2), the following equation is obtained:

$$J = \frac{Eit}{(M_s + W)(\theta_2 - \theta_1)} \frac{\text{joule}}{\text{calories}} \quad (10.3)$$

10.4 Experimental Methods

10.4.1 Procedure

1. Weigh the empty calorimeter along with the stirrer (inner vessel). Then fill it with water until the heating wire is approximately submerged, and weigh the entire system. This is recorded as the total mass. The mass of the water (M) is the difference between these two measurements.
2. Set-up the calorimeter.
3. Assemble the electrical circuit as shown in the diagram above. Do not connect it to the power source before having it inspected by the assistant.
4. After inspection, connect it to the power source, close the switch, and adjust the sliding resistor so that the current is approximately 3–5 amperes. Ensure the heater is submerged in water before closing the switch. Then open the switch.
5. Stir the water gently and record the initial temperature (T_m). Measure and record the temperature at one-minute intervals for 10 minutes. Also record the current and voltage at each time interval.
6. Close the switch when the clock's second-hand points to zero. Note the time, then record the potential difference and current every minute to monitor for any fluctuations in voltage. Also record the thermometer reading. Stir the water in the calorimeter gently throughout the observation.
7. After 10 minutes, turn off the electrical circuit. Continue recording the temperature at one-minute

intervals for another 10 minutes, while gently stirring the vessel.

8. Determine the maximum and final temperatures. These will be used to calculate the radiation correction.
9. Calculate the water equivalent of the calorimeter.
10. Repeat the experiment using a different mass of water.
The experiment only requires two different water masses.

10.4.2 Recorded Experimental Data

Mass of calorimeter + stirrer	$m_1 =$	 g
Mass of calorimeter + stirrer + water	$m_2 =$	 g
Mass of water	$M = m_2 - m_1 =$	 g
Specific heat capacity	$s =$	1 kal. g ⁻¹ . °C ⁻¹
Specific heat capacity of metal	$s_l =$	... kal. g ⁻¹ . °C ⁻¹
Water equivalent of the calorimeter	$W = m_l s_l =$	 g
Initial temperature of calorimeter + water	$\theta_i =$	 °C
Maximum temperature + water	$\theta_m =$	 °C
Final temperature of calorimeter + water	$\theta_f =$	 °C
Temperature increase	$(\theta_m - \theta_i) =$	 °C
Radiation correction	$(\theta_m - \theta_f)/2 =$	 °C
Corrected temperature increase	$(\theta_2 - \theta_1)$ $= (\theta + \theta_2 =$	 °C
Time during current flow	t	 sec
Average current during time interval t	i	 A
Average voltage during time interval t	E	 V

Heat-electricity equivalence

$$J = \frac{Eit}{(M_s + W)(\theta_2 - \theta_1)} \frac{\text{joule}}{\text{calories}}$$